MIL-V-5635A
26 April 1966
Superseding
MIL-V-5635
9 February 1955

MILITARY SPECIFICATION

VIBRATOR, IGNITION COIL, AIRCRAFT ENGINE STARTING, TYPE AN4181

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a magnetically operated, radio interference suppressed starting vibrator for use with aircraft engine magnetos.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating; Cadmium (Electrodeposited)
QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (for Use with Electronic Equipment in Aircraft)
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-E-5557	Enamel; Heat Resisting, Glyceryl-Phthalate, Black
MIL-S-7742	Screw Threads, Standard Optimum Selected Series: General Specification for
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

STANDARDS

Military

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS33540	Safety Wiring, General Practices for
MS33586	Metals, Definition of Dissimilar

Air Force-Navy Aeronautical

AN4181	Vibrator, Aircraft Engine Starting, Radio-Interference Suppressed

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provision for preproduction testing.

3.2 Selection of specifications & standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.2.1 AN or MIL standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part number. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they develop suitable properties and are replaceable by AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced in the parts list and if practicable, on the contractor's drawings. In case there is no suitable corresponding AN or MIL standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Materials. Materials used in the manufacture of starting vibrators shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to contractors' specifications may be used provided the specifications are released by the Government and contain provisions for adequate tests. The use of contractors' specifications will not constitute waiver of Government inspection.

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3.3.1 Metals. Metals shall be of the corrosion resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4 Design and construction. The starting vibrator shall be designed to furnish a continuous flow of sparks when properly connected to an aircraft magneto. A single assembly shall be constructed of two units: A vibrating circuit interrupter and a relay switch enclosed within a housing. A multiple assembly may be made for use with more than one magneto which shall consist of the one vibrating circuit interrupter for each magneto and a total of one or more relay switches within a housing. The design and the construction shall readily permit repair or replacement of internal parts with a minimum of disassembly.

3.4.1 Vibrating circuit interrupter. The vibrating interrupter shall supply pulsating direct current of sufficient frequency to induce the required output voltage in the magneto.

3.4.2 Relay switch. The relay switch shall automatically connect and disconnect the starting vibrator from the magneto primary grounding connection. When the starting vibrator is not operating, the circuit between the vibrating circuit interrupter and the magneto shall be open. The relay switch contact connecting to the vibrating circuit interrupter shall be grounded when in the deenergized position.

3.4.3 Housing. The housing shall have the number and type of terminals for shielded conductors as specified in AN4181. The housing, cover, and base shall provide a complete electrical shield and the housing shall also be fire and explosion proof. In addition, provision shall be made by providing sufficient ventilation and by other means to prevent corrosion and condensation. Suitable means shall be provided for bonding the vibrator housing to the aircraft. If shock isolators are used, bonding shall be in accordance with MIL-C-172.

3.4.4 Cover. The starting vibrator shall be provided with an easily removable cover for inspection and adjustment.

3.4.5 Polarity and voltage. Unless otherwise specified, the starting vibrator shall be designed to operate only in 24-volt electrical system where the negative pole of the battery is grounded.

3.4.6 Threaded parts.

3.4.6.1 Screw threads. Screw threads shall be in accordance with MIL-S-7742, Class 2 fit.

3.4.6.2 Locking. All internal or external threaded parts shall be positively locked or safety wired in accordance with MS33540.

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3.5 Performance. The starting vibrator shall satisfy the performance requirements specified in Section 4, when subjected to the applicable tests.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of MIL-D-1000.

3.7 Radio interference. The starting vibrator shall satisfy the radio interference test requirements specified in Section 4, when subjected to the applicable tests.

3.8 Dimensions. The dimensions shall conform to AN4181.

3.9 Weight. The weight including ground terminals shall not exceed the value specified in AN4181.

3.10 Finish.

3.10.1 Plating. Steel parts in contact with aluminum or aluminum alloy shall be cadmium-plated in accordance with QQ-P-416 or zincplated in accordance with QQ-Z-325.

3.10.2 Anodizing. All aluminum-alloy parts shall be anodized in accordance with MIL-A-8625.

3.10.3 Paint finish. All external surfaces, except anodized aluminum alloy, corrosion-resisting steel, working surfaces, threads, and cadmium- and zinc-plated parts, shall be finished with one coat of primer conforming to MIL-P-8585 and two coats of black enamel conforming to MIL-E-5557.

3.11 Operation markings.

3.11.1 Terminals. The terminals shall be legibly die stamped as specified in AN 4181.

3.12 Product marking.

3.12.1 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13 Workmanship. All details of workmanship shall be in accordance with high grade electrical ignition manufacturing practice common to this type of equipment.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspec-

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tions set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the starting vibrator shall be classified as follows:

a. Preproduction tests.
b. Acceptance tests.

4.3 Preproduction testing.

4.3.1 Test sample. The preproduction test sample shall consist of six starting vibrators representative of the production vibrator. The sample shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity. (see 6.3)

4.3.2 Test report. Upon completion of preproduction testing, the contractor shall submit a test report prepared in accordance with MIL-STD-831.

4.3.3 Preproduction tests. The preproduction tests of sample starting vibrators in the quantity specified shall consist of all the acceptance tests and the following tests. The preproduction tests may, at the option of the procuring activity, be supplemented with tests under actual service conditions to determine conformance with this specification.

4.3.3.1 Intermittent service test. One starting vibrator shall be subjected to an Intermittent test as follows:

One cycle of 6 minutes on and one-half hour off.
Forty-eight hours of cycles consisting of 1 minute on and 6 minutes off.

This test shall be conducted at room temperature, and during the "on cycle" 27 volts (V_1) shall be applied.

4.3.3.2 Vibration.

4.3.3.2.1 Vibration survey. Three starting vibrators shall be subjected to this survey. Each vibrator shall be mounted on a Vibration test stand and subjected to linear vibration over the range of double amplitudes and frequencies indicated by the curve of figure 1. The direction of motion shall be parallel to that of the relay armature contact point. The rate of change of the frequency of the applied vibration shall be sufficiently slow to permit detection of any closing of the normally open contact points by means of an external circuit, incorporating a suitable continuity indicator. Any indication of continuity resulting from closing of the relay points shall constitute rejection. During this test the vibrators shall be nonfunctioning.

4.3.3.2.2 Vibration endurance. The three starting vibrators used during the vibration survey shall be subjected to this test at the same time, one in each

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of the positions specified. The starting vibrators shall be connected as shown in figure 2 and in addition shall be mounted on a vibration test stand which shall vibrate at a frequency of 60 cycles per second through an amplitude of .010 inches in a horizontal plane. One starting vibrator shall be mounted horizontally with the base down, one shall be mounted vertically with the axis of the terminals horizontal, and one shall be mounted on a plane inclined 45 degrees to the horizontal with the base down and the axis of the terminals horizontal. The starting vibrators shall be vibrated for 100 hours. The starting vibrator shall be subjected to the magneto voltage output nominal input test, after 50 hours and at the end of the test and shall maintain the specified voltage output without adjustment.

4.3.3.3 Remaining tests. One of the three vibrators subjected to the Vibration test shall be subjected to the following tests in the order specified.

4.3.3.3.1 High temperature operation. The starting vibrator shall be heated at a temperature of 93°C (200°F) for 2 hours. At this temperature, it shall be subjected to and pass all the acceptance tests.

4.3.3.3.2 Low temperature operation. The starting vibrator shall be placed in a cold room and after 2 hours at an ambient of -54°C (-65°F) shall be subjected to and pass all the acceptance tests.

4.3.3.3.3 Magneto voltage output. This test shall be conducted using each of the magneto models specified by the manufacturer. The open circuit voltage output (V_2) on two leads, resulting from the action of two consecutive lobes of the breaker cam, shall be measured by a peak voltmeter with the magneto rotor stationary.

4.3.3.3.3.1 High input. Thirty volts (V_1) shall be applied to the starting vibrator for 5 minutes. The magneto output voltage (V_2) shall be measured at the beginning and end of the period. The peak voltage output of either lead shall not be less than 10 kv.

4.3.3.3.3.2 Low input. Thirteen volts (V_1) shall be applied to the starting vibrator and the magneto output voltage (V_2) measured. The peak voltage output of either lead shall not be less than 10 kv.

4.3.3.4 Radio interference test.

4.3.3.4.1 Test set-up. The radio interference test set-up shall conform to figures 3 and 4. The following explanations apply to these figures:

a. Signal generators, receivers, and output meters used shall be of such accuracy, sensitivity, shielding, and output that will give reliable measurements at the specified levels.
b. The 50 ohms, 6 db, attenuation pad shall be Measurements Corporation part No. 80-ZH3 or equal, and it shall be matched to its respective signal generator or receiver.
c. Co-axial switches. Bird Electronic Corporation, Mod. 74, or equal.
d. All connectors shall be of 50 ohms Type N construction.

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e. Receiver Matching Network =

.55 - 18mc	100 mfd in series
18 - 140mc	75 ohms in series
Above - 500mc	50 ohms, 6 db pad as in (b) above.

f. Switch positions for tests as shown in the circuit diagram.

Calibrate position - Both in position A
Specimen position - Both in position B
Neutral position - Signal generator switch in position B, receiver switch in position C (neutral)

g. Signal generator, receiver, test vibrator and co-axial switches are to be mounted and ground to a common copper ground plane so that no disturbing stray fields exist.

h. There should be no indication of leakage upon the output meter when both co-axial switches are in position C.

i. When the signal generator is de-energized the receiver should not pick up any signal, regardless of the positioning of the co-axial switch. In order to satisfy this condition, it may be necessary to use a screen room.

j. The grounding of the test vibrator referred to in (g) above shall be accomplished by means of the bonding provisions furnished.

4.3.3.4.2 Attenuation. The starting vibrator shall have an attenuation ratio of 70 db in the frequency range from 0.550 to 140 mcs and 50 db from 500 to 1000 mcs between the magneto and battery terminals. This requirement shall be achieved by proper design to reduce capacity coupling rather than by the incorporation of filters. The attenuation of the starting vibrator is:

$$20 \text{ Log } \frac{\text{Attenuator setting on specimen position}}{10 \text{ Attenuator setting on calibrate position}}$$

Where the signal generator attenuator is adjusted to give the same receiver output reading in each position.

4.3.3.4.3 Case radiation. Case radiation attenuation ratio shall be at least 90 db in the frequency range from 100 to 140 mcs. This measurement is made in the same manner as the test in the paragraph entitled, "Attenuation," except that the receiver co-axial switch is placed in the neutral position (disconnected from the direct lead and the test specimen). The case radiation attenuation of the starting vibrator is:

$$20 \text{ Log } \frac{\text{Attenuator setting in Neutral position}}{10 \text{ Attenuator setting on calibrate position}}$$

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Where the signal generator attenuator is adjusted to give the same receiver output reading in each position.

4.4 Acceptance tests. When inspection is conducted at the contractor's plant, the tests shall be accomplished by the contractor under the supervision of the Government Inspector. Contractors not having laboratory facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in this specification. Previous acceptance or approval of material, or the release of any design by the Government, shall not be construed as a guaranty of the acceptance of the finished product.

4.4.1 Test set-up. Unless otherwise specified, the set-up for acceptance tests, shall conform to figure 2. The magneto shall operate at 20-40 magneto rpm.

4.4.2 Individual tests. The following tests shall be conducted on each starting vibrator submitted for acceptance under contract.

4.4.2.1 Examination of product. Each starting vibrator shall be examined to determine conformance with this specification with respect to materials, workmanship, marking and the applicable drawings.

4.4.2.2 Relay switch cut-in voltage. The switch to the rheostat shall be closed and the voltage (V_1) lowered to nearly zero. The voltage shall then be raised until the points of the relay switch close. The points shall close at a maximum of 13 volts.

4.4.2.3 Magneto voltage output, nominal input. For preproduction this test shall be conducted using the magneto models for which the vibrator has been designed. For the acceptance tests, the magneto used in the test set-up shall be that magneto on the specified list having the lowest output. Twenty-seven volts (V_1) shall be applied to the starting vibrator and the output voltages (V_2) on two leads, resulting from the action of two consecutive lobes of the breaker cam, shall be measured by a peak voltmeter with the magneto rotor stationary. The peak voltage of either lead shall not be less than 10 kv.

4.4.2.4 Current consumption. The current consumption in amperes (A_1) shall be measured during the magneto voltage output, nominal input test, and shall not exceed the amperage rating by more than 0.5 amperes per unit.

4.4.2.5 Insulation. All insulated parts of the vibrator shall be subjected to and shall withstand 1000 volts direct current for 1 minute without damage.

4.4.3 Rejection and retest. Any starting vibrator failing to meet the requirements of the specified tests shall be rejected and returned at the contractor's expense. Starting vibrators which have been rejected may be replaced or repaired to correct the defects and resubmitted for all specified tests. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the original defects shall be furnished to the inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Procuring Service.

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4.5 Inspection of the preservation, packaging and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order, each vibrator shall be individually preserved and packaged in accordance with MIL-P-116, Method IIC, overboxed in a unit container conforming to class Domestic, Type 275, PPP-B-636.

5.1.2 Level C. Unless otherwise specified in the contract or order, vibrators shall be individually preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

5.2.1 Level A. Vibrators preserved and packaged as specified in 5.1.1 shall be packed in export type shipping containers conforming to PPP-B-636. As far as practicable, exterior shipping containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Containers shall be closed and strapped in accordance with the specification and applicable appendix thereto.

5.2.2 Level B. Vibrators preserved and packaged as specified in 5.1.1 shall be packed in domestic type exterior shipping containers conforming to PPP-B-636. Exterior shipping containers shall be of minimum cube and tare consistent with the protection required. As far as practicable, exterior shipping containers shall be of uniform shape and size and contain identical quantities. The gross weight of each shipping container shall not exceed the weight limitation of the container specification. Containers shall be closed and strapped in accordance with the specification and applicable appendix thereto. Shipping containers shall meet the special requirements table of PPP-B-636, as applicable.

5.2.3 Level C. Packages that require over-packing for acceptance by the carrier shall be placed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Physical protection. Cushioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186, except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of MIL-STD-1186 shall be waived when preservation, packaging, and packing of the item is for immediate use or when drop tests of MIL-P-116 are applicable.

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5.4 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

"VIBRATOR-AIRCRAFT ENGINE STARTING"
AN designation
Specification MIL-V-5635A
Manufacturer's part number
Name of manufacturer
Name of contractor (if different from manufacturer)
Contract or Order No.

6. NOTES

6.1 Intended use. The starting vibrator covered in this specification is intended for use with aircraft engine magnetos to provide a flow of sparks at the spark plugs while starting the engine.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.
b. Applicable levels of preservation, packaging and packing required (See 5.1 and 5.2).

6.3 Instructions for preproduction sample. Size and testing site (see 4.3.1).

6.4 Definitions.

6.4.1 Amperage rating. The amperage rating shall be the starting vibrator current consumption established by the manufacturer and measured with an ammeter at normal input voltage of 27 volts.

6.4.2 Voltage rating. The voltage rating shall be the input voltage of 27 volts.

6.5 Identification of changes. Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the changes.

Custodians:
Army - MO
Navy - WP
Air Force - 82

Review activities:
Army - MO
Air Force - 82

Preparing activity:
Air Force - 82

Project No. 2925-0016

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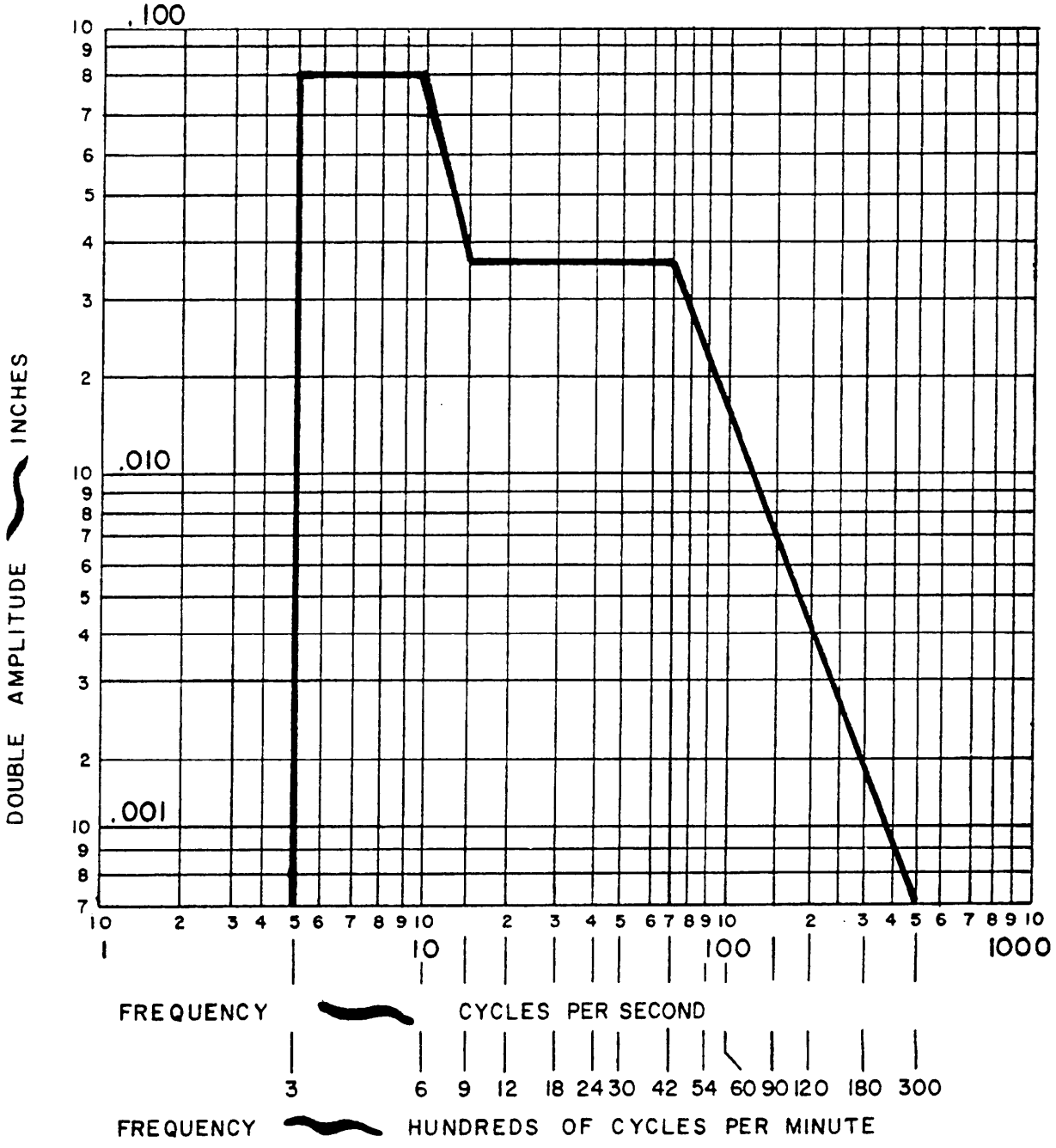


FIGURE 1. Range Curve for Vibration Survey

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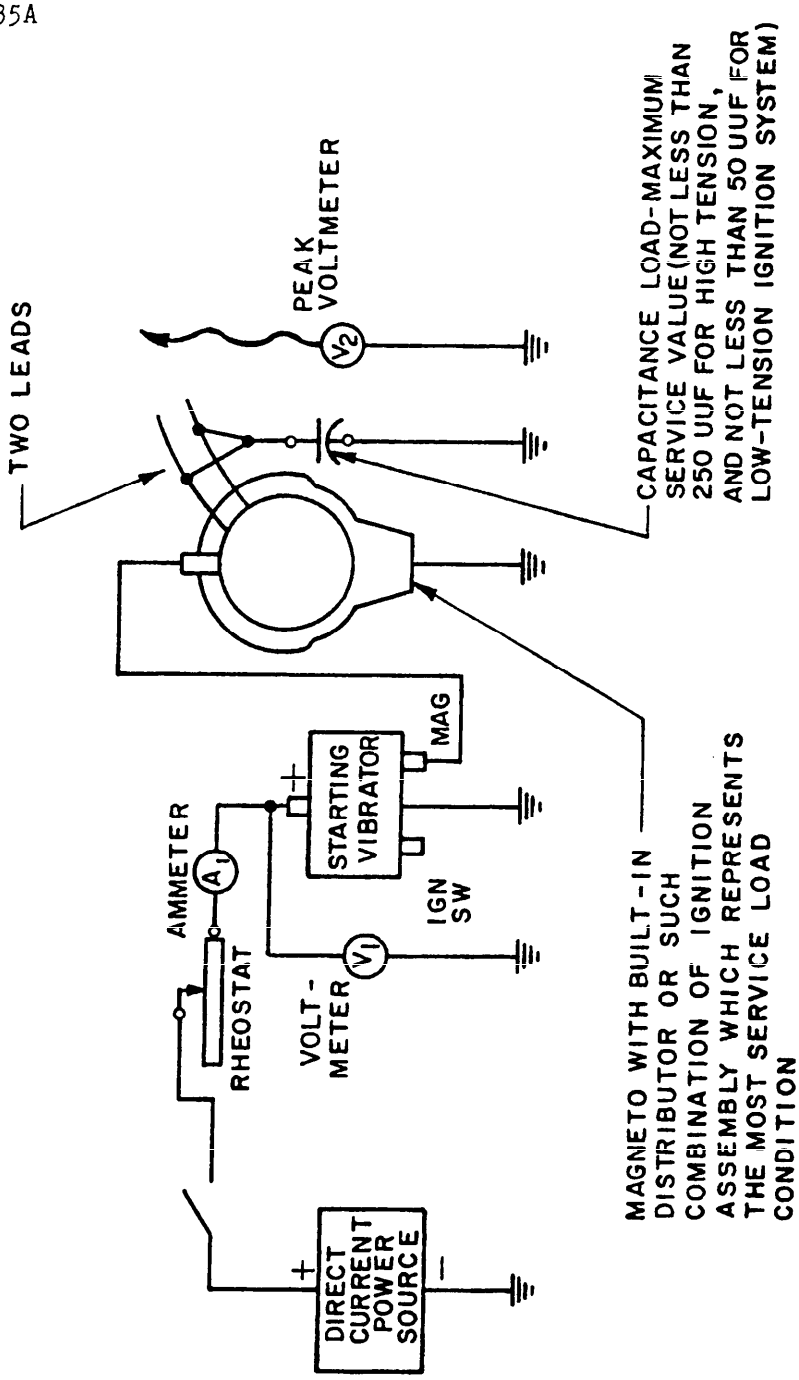


FIGURE 2. Test Set-Up

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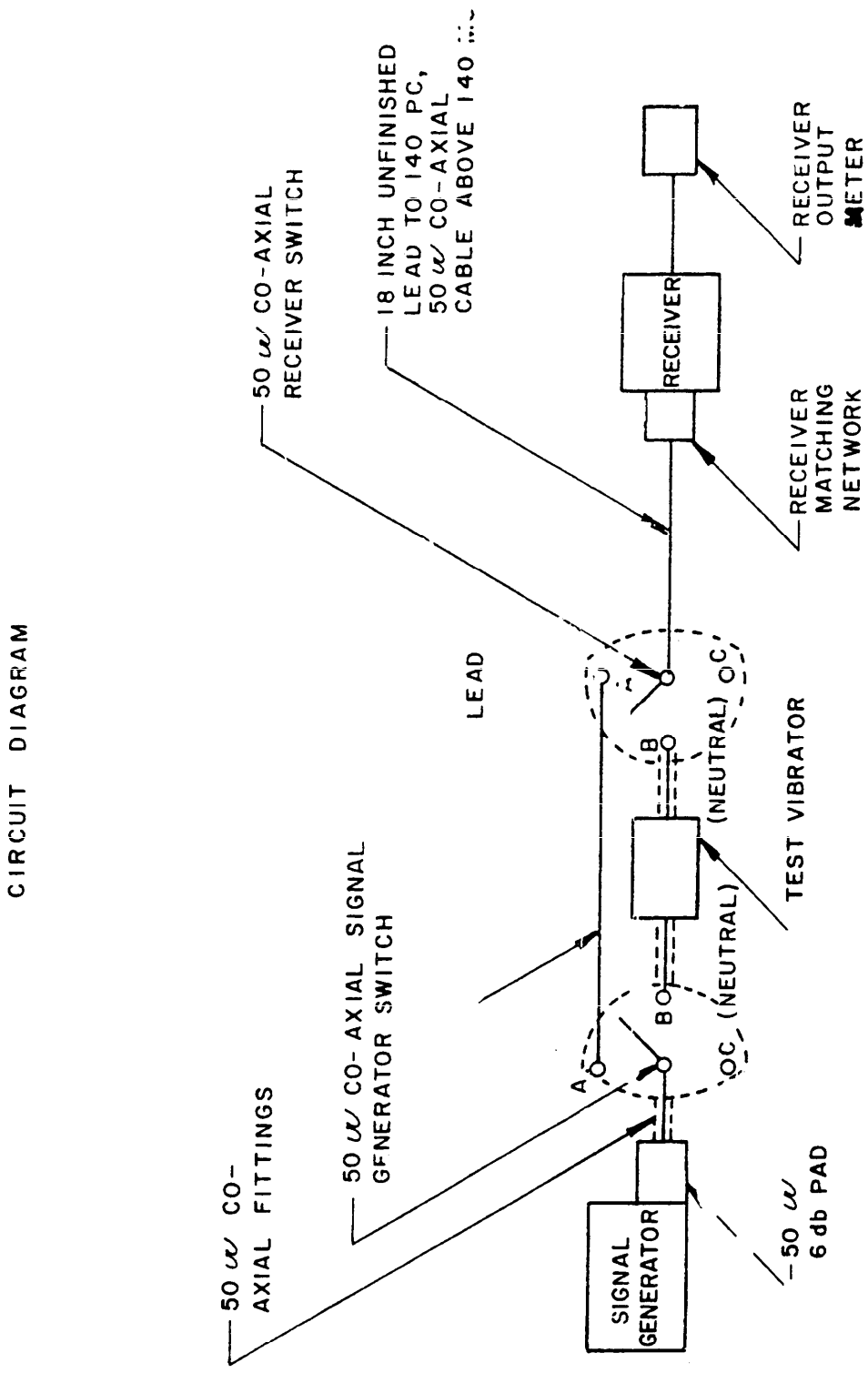


FIGURE 3. Radio Interference Test Set-Up (Circuit Diagram)

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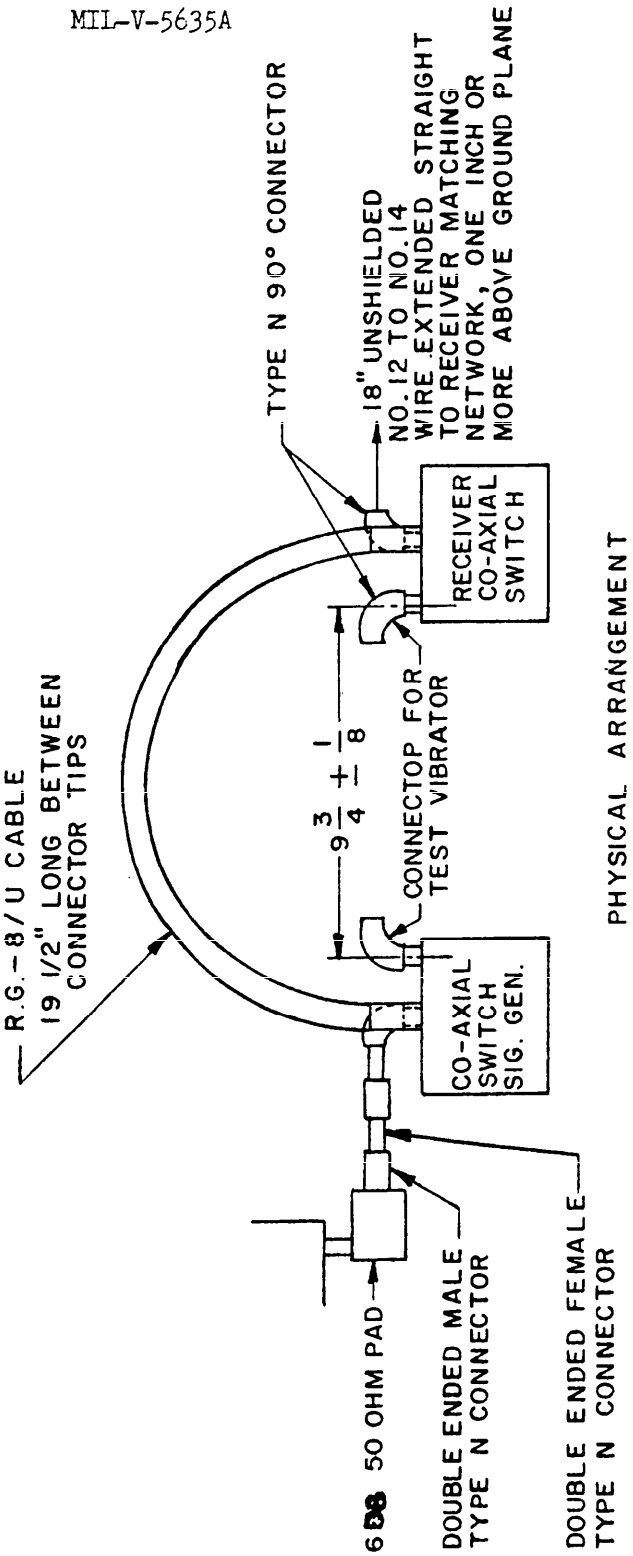


FIGURE 4. Radio Interference Test Set-Up (Physical Arrangement)

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

D Form 1426

Figure V-5, Example of Specification Analysis Sheet (front)

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KELLY AFB, TEXAS 78241

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Figure V-5, Example of Specification Analysis Sheet (back)

AFPS SA MAY 64, 500

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